

Han Yan

Email: k4rtol@163.com | <https://k4rto.com/>

Education

Australian National University (ANU) QS #34 — Master of Computing · ML Specialization

Key courses: Computer Vision, Computer Graphics, Game Development, (Statistical) Machine Learning, Advanced Algorithms, SQL Databases.

Donghua University (211) — Bachelor of Functional Materials

Internships & Work Experience

Hichains | ML / AI Researcher

2025.04 – Present

- **Smart OCR Hub — Lenovo Customs Declaration Auto-Recognition**
 - **End-to-end OCR pipeline:** Vision-Language Model primary path with LLM field merging; on-prem CV model as fallback for resilience when the upstream API is unreachable.
 - **Multi-GPU inference cluster:** real-time scheduling + GPU utilisation / alert monitoring, supporting batch-concurrent multi-document processing.
 - **End-to-end MLOps:** experiment tracking + human-review loop + continuous fine-tuning, giving every model iteration an audit trail.
 - **Stack:** Python · FastAPI · vLLM · Redis · PostgreSQL · MLflow · SvelteKit.
- **AURA — Enterprise-grade Multi-Agent Collaboration Platform**
 - **Blueprint orchestration engine:** visual DAG editor turns SOPs into reusable agent workflows with parallel dispatch, rate-limiting, version snapshots, and replay.
 - **RAG + self-hosted private data:** retrieval-augmented Q&A over a private knowledge base, deployed inside the customer's perimeter — data never leaves the org.
 - **Fine-tuning & post-training:** built-in training pipeline so business-specific models accrete and continuously evolve inside the organisation.
 - **Real-time collaboration core:** WebSocket + CRDT collaborative editing, supporting multi-user multi-Agent parallel tasks on a single screen.
 - **Stack:** Python · FastAPI · SQLAlchemy 2.0 · PostgreSQL · SvelteKit 5 · MLflow.

ANU Techlauncher Capstone | Vision Engineer + PM

2024.02 – 2024.11

- **Human Fall Pose Detection & Audio Alert System (Python) — Joint project with a Western Australia company**
 - **Vision algorithm development:** customized YOLOv7 model for real-time human keypoint detection and pose analysis; structural optimization for accurate fall recognition.
 - **Project Management (PM):** led project planning and module assignment, using Jira for team task tracking and progress visibility.

Highlight: Presented at the ANU Techlauncher showcase.

Details: [CAM FALL Project](#)